HEALTHCARE IoT NETWORK SECURITY & INTRUSION DETECTION SYSTEM

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THE PROBLEM

Healthcare Under Attack

- 82% of hospitals hit by cyberattacks
- $10.9M average cost per breach
- Ransomware shuts down critical care
- Medical devices have unpatched vulnerabilities

Real Consequences

- 2020: 400 hospitals shut down (ransomware)
- 2021: 465,000 pacemakers recalled (hacking risk)
- 2022: Insulin pumps remotely hacked
- 2023: Surgical units shut down (IoT botnet)

OUR SOLUTION

We built a complete hospital network simulation demonstrating both VULNERABLE and SECURE IoT architectures:

1	**Simulated Hospital Network:**	7 Raspberry Pi devices — 5 simulate cardiac patients (ECG), 2 control the physical attack demo
2	**Enterprise Security Infrastructure:**	Industry-standard tools: firewalls, intrusion detection, and security monitoring
3	**Live Attack Demonstration:**	Physical proof: vulnerable system fails catastrophically, secure system defends successfully

SYSTEM OVERVIEW

IDS = Intrusion Detection System • SIEM = Security Information & Event Management

7 RASPBERRY Pi 5 "PATIENTS" + DEMO — 5 cardiac ECG simulators + 2 attack demo controllers

DATA PROCESSING PIPELINE — Real-time collection → Database → Dashboards

SECURITY MONITORING
Firewall • IDS • SIEM

ATTACK DEMO
Vulnerable vs. Secure System

BY THE NUMBERS

100	9	14	53M	100%	0%
readings/sec	virtual servers	security agents	data points	attack success	attack success
Data Pipeline	Infrastructure	Monitoring	Collected	Vulnerable System	Secure System

MULTI-LAYER SECURITY

7	**Physical Validation**	Sensor cross-checking against ground truth
6	**Data Encryption**	SSL/TLS, encrypted databases
5	**Secrets Management**	HashiCorp Vault — credentials & keys
4	**Identity & Access Control**	Keycloak SSO — centralized authentication
3	**Security Monitoring**	Wazuh SIEM — 14 agents across all systems
2	**Intrusion Detection**	Suricata IDS — real-time network analysis
1	**Network Segmentation**	Firewall + VLANs — isolated threat zones

KEY TECHNOLOGIES

Hardware

- Raspberry Pi 5 (7 units) — 5 patient ECG simulators, 2 demo controllers
- Pressure sensors & pneumatic control systems
- Network switches, UniFi firewall (UDM-SE)

Software Infrastructure

- PostgreSQL — database (100 inserts/sec)
- Grafana — real-time visualization dashboards
- Node-RED — IoT data flow & routing

Security Tools

- Suricata — network intrusion detection (IDS)
- Wazuh — security monitoring & alerts (SIEM)
- Keycloak — authentication & access control
- HashiCorp Vault — secrets management

Standards & Compliance

- HL7 FHIR — healthcare data interoperability standard
- HIPAA — compliance controls & audit trails